

LAB EXPERIMENT RECORD

Set 4: Product Inventory Management

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DIVYASREE.M
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Aim

To visualize product inventory data using a bar chart, stacked bar chart and scatter plot, and to build an interactive dashboard combining the bar chart and stacked bar chart.

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Procedure

1. Create vectors for product name and quantity available.
2. Use `barplot()` to visualize the quantity of each product.
3. Create a category vector, build a quantity table with `xtabs()`, and draw the stacked bar chart with `barplot()` and `legend()`.
4. Create a price vector and use `plot()`, `lm()` and `cor()` to analyse price versus quantity.
5. Build a Shiny dashboard with category and minimum-quantity filters showing the bar and stacked bar charts together, and run it with `shinyApp()`.

Key R Functions Used

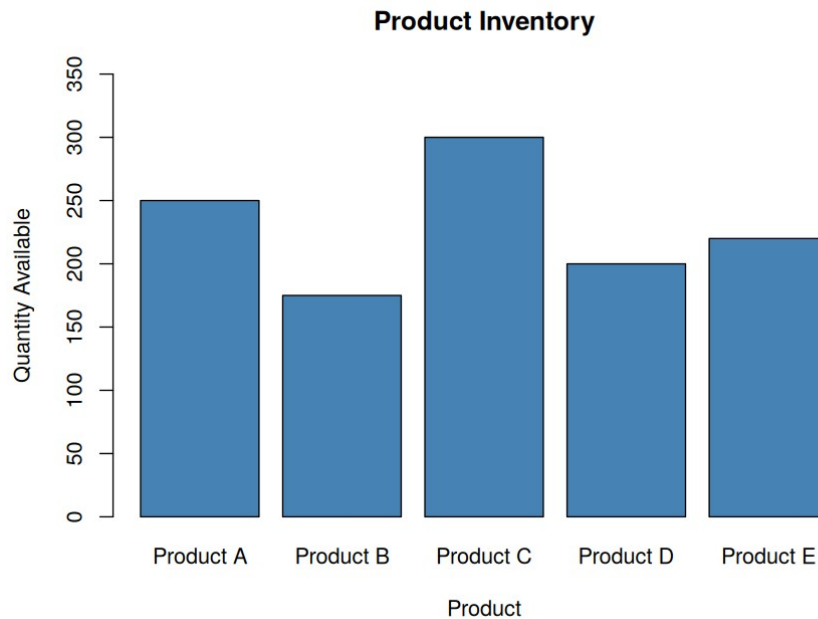
Function	Purpose
<code>barplot()</code> / <code>legend()</code>	Create bar and stacked bar charts with a legend.
<code>xtabs()</code>	Builds a quantity table by product and category.
<code>plot()</code> / <code>text()</code>	Create the scatter plot and label points.
<code>lm()</code> / <code>abline()</code> / <code>cor()</code>	Fit the trend line and measure correlation.
<code>selectInput()</code> / <code>numericInput()</code>	Provide dashboard filters.
<code>reactive()</code> / <code>renderPlot()</code> / <code>shinyApp()</code>	Update charts dynamically and run the app.

PROGRAM 1: Bar Chart of Product Inventory

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Dataset
product_name <- c("Product A", "Product B", "Product C",
                  "Product D", "Product E")
quantity      <- c(250, 175, 300, 200, 220)

# Bar Chart
barplot(quantity,
        names.arg = product_name,
        col = "steelblue",
        ylim = c(0, 350),
        xlab = "Product",
        ylab = "Quantity Available",
        main = "Product Inventory")
```



Output 1: Product Inventory (Bar Chart)

Observation: Product C has the highest stock (300 units) and Product B the lowest (175 units). Total inventory is 1,145 units, an average of 229 units per product.

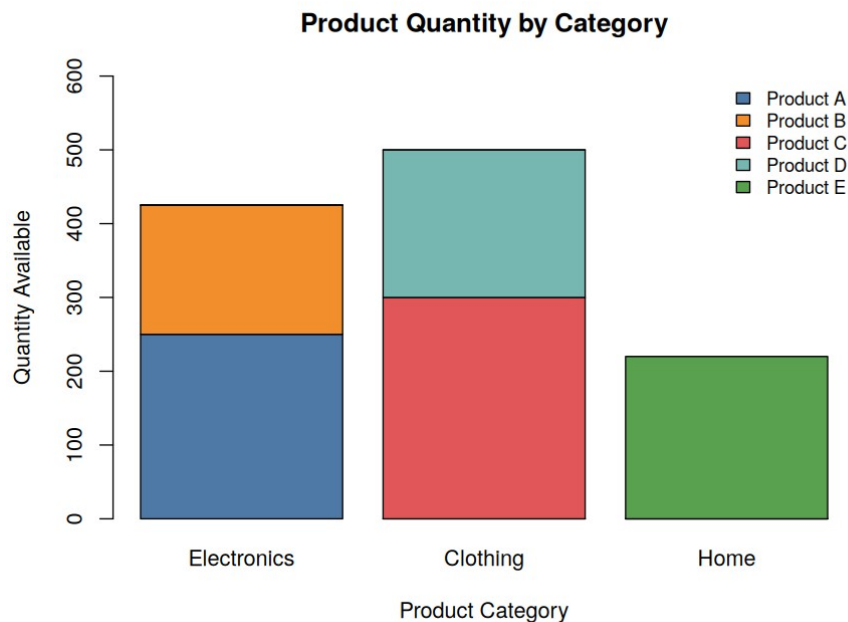
PROGRAM 2: Stacked Bar Chart of Product Quantity by Category

Generate a stacked bar chart to show the quantity of each product within different product categories.

```
# Dataset
product_name <- c("Product A", "Product B", "Product C",
                  "Product D", "Product E")
quantity     <- c(250, 175, 300, 200, 220)
category     <- c("Electronics", "Electronics", "Clothing",
                  "Clothing", "Home")

# Quantity of each product within each category
inv <- xtabs(quantity ~ product_name + category)
inv <- inv[, c("Electronics", "Clothing", "Home")]
colours <- c("#4E79A7", "#F28E2B", "#E15759", "#76B7B2", "#59A14F")

# Stacked Bar Chart
barplot(inv,
        col = colours,
        ylim = c(0, 600),
        xlab = "Product Category",
        ylab = "Quantity Available",
        main = "Product Quantity by Category")
legend("topright", legend = rownames(inv), fill = colours, bty = "n")
```



Output 2: Product Quantity by Category (Stacked Bar Chart)

Observation: Clothing holds the most stock (500 units: C 300 + D 200), followed by Electronics (425 units: A 250 + B 175) and Home (220 units: E). Each bar is split by product, so both the category total and each product's share are visible.

Note: The dataset has no category column, so example categories (Electronics, Clothing, Home) were assumed.

PROGRAM 3: Scatter Plot of Product Price vs Quantity

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
# Dataset
product_name <- c("Product A", "Product B", "Product C",
                  "Product D", "Product E")
quantity      <- c(250, 175, 300, 200, 220)
price         <- c(20, 15, 18, 22, 17)

# Scatter Plot
plot(price, quantity,
     pch = 19,
     col = "purple",
     cex = 1.4,
     xlim = c(14, 23),
     xlab = "Product Price ($)",
     ylab = "Quantity Available",
     main = "Product Price vs Quantity Available")
text(price, quantity, labels = product_name, pos = 4, cex = 0.75)
abline(lm(quantity ~ price), col = "gray40", lty = 2)

# Correlation
cor(price, quantity)      # 0.19
```



Output 3: Product Price vs Quantity Available (Scatter Plot)

Findings: The correlation is 0.19, a very weak positive relationship, so price has little effect on stock level. Product C (\$18) has the most stock and Product B (\$15, the cheapest) the least, while the most expensive Product D (\$22) has a middle stock level of 200.

Note: The dataset has no price column, so example price values were assumed.

PROGRAM 4: Interactive Inventory Dashboard

Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

```
library(shiny)

# Dataset
inventory <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Electronics", "Clothing", "Clothing", "Home"),
  Quantity = c(250, 175, 300, 200, 220)
)
prod_cols <- c("Product A" = "#4E79A7", "Product B" = "#F28E2B", "Product C" = "#E15759",
               "Product D" = "#76B7B2", "Product E" = "#59A14F")

ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      selectInput("category", "Select Categories:",
        choices = unique(inventory$Category),
        selected = unique(inventory$Category),
        multiple = TRUE),
      numericInput("minqty", "Minimum Quantity:",
        value = 0, min = 0, max = 300, step = 25),
      hr(),
      strong("Summary"),
      verbatimTextOutput("summary")
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("barPlot", height = "350px")),
        column(6, plotOutput("stackedPlot", height = "350px"))
      )
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({
    inventory[inventory$Category %in% input$category &
              inventory$Quantity >= input$minqty, ]
  })

  output$barPlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0)
    barplot(d$Quantity, names.arg = d$Product, col = "steelblue",
            cex.names = 0.8, ylim = c(0, 350),
            xlab = "Product", ylab = "Quantity Available",
            main = "Product Inventory")
  }, width = 420, height = 340)

  output$stackedPlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0)
    tab <- xtabs(Quantity ~ Product + Category, data = d)
    barplot(tab, col = prod_cols[rownames(tab)], ylim = c(0, 600),
            xlab = "Product Category", ylab = "Quantity Available",
            main = "Product Category")
    legend("topright", legend = rownames(tab), fill = prod_cols[rownames(tab)],
           bty = "n", cex = 0.8)
  })
}
```

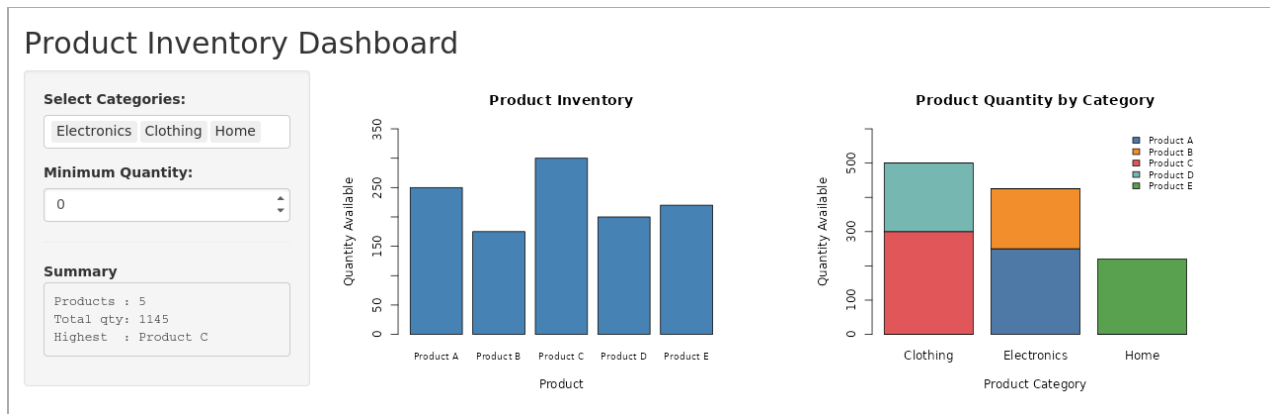
```

}, width = 420, height = 340)

output$summary <- renderText({
  d <- filtered()
  req(nrow(d) > 0)
  paste0("Products : ", nrow(d), "\n",
        "Total qty: ", sum(d$Quantity), "\n",
        "Highest : ", d$Product[which.max(d$Quantity)])
})
}

shinyApp(ui = ui, server = server)

```



Output 4: Interactive Product Inventory Dashboard (R Shiny)

Observation: The dashboard shows the product bar chart and the category stacked bar chart side by side. Selecting categories or setting a minimum quantity updates both charts and the summary — with all products, total stock is 1,145 units and Product C is the highest.

Note: The question specifies Tableau; since this record uses R, the equivalent dashboard was built with Shiny. In Tableau: connect to the data, create a bar chart sheet (Product Name vs Quantity), create a stacked bar sheet (Category on Columns, Quantity on Rows, Product Name on Color), place both on a dashboard, and add Category as a filter applied to both sheets.

Result

The product inventory data was successfully visualized using a bar chart, stacked bar chart and scatter plot, and an interactive dashboard combining the bar and stacked bar charts was created.